WARNING Be cautious when using infinite generators. Always ensure you have a proper termination condition to avoid infinite loops or memory leaks.

Now let's see how this `async` iterable can be consumed using a `for await...of` loop in the following listing.

Listing 3.5 Consuming an `async` iterable

```
(async () => {
  for await (const number of getOneToThree()) {.
    console.log(number);
  }
})();
```

Uses the for await...of loop to yield the next number in the sequence

Table 3.2 shows a step-by-step visualization of how the `async` iterable is processed. The `async` iterable allows for asynchronous generation and consumption of values. The `for await...of` loop waits for each value to be yielded by the `async` iterable before proceeding to the next iteration. This enables the processing of values as they become available, without blocking the execution of other code at the same time. The same process of consuming the stream of values is used when consuming a streaming text response that originates from an AI model.

Table 3.2 What happens on each iteration of the `for await ...of` loop on an `async` iterable function

Time	Yield	`for await...of` loop
0	Start	`Start`
1	Yield 1	`console.log(1)`
2	Yield 2	`console.log(2)`
3	Yield 3	`console.log(3)`
4	End	`End`

Now let's go back to our example: other than the `textStream` response field, the `streamText` function also provides additional features and callbacks represented in table 3.3.

Table 3.3 Features and result helper functions of the `streamText` utility

Feature	Comment
`onFinish` callback	You can provide a callback that is triggered when the model finishes generating the response and all tool executions. This callback receives the generated text, tool calls, tool results, finish reason, and usage information.
Result helper functions	The result object returned by `streamText` contains helper functions like `toAIStream()`, `toAIStreamResponse()`, `toTextStreamResponse()`, `pipeTextStreamToResponse()`, and `pipeAIStreamToResponse()` to facilitate integration with AI SDK UI components and streaming responses.